



SIMATS ENGINEERING

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Course Code: CSA-1522

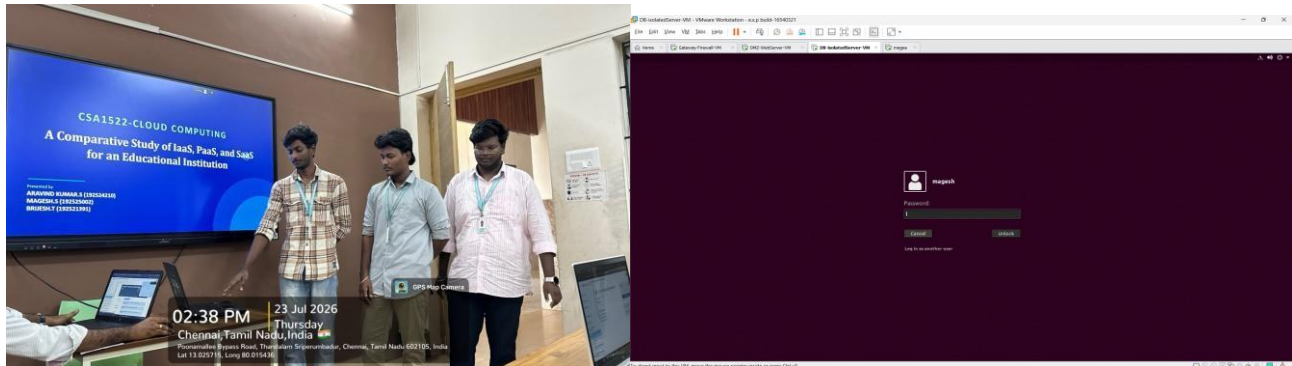
Slot: C

Course Name: Cloud Computing and Big Data Analytics

Course Faculty: Dr. P. Rajaram

Project Title: Secure Multi-VM Isolation and Threat Detection System in Virtualized Environments

Module Photographs:



Project Description:

This project is a Secure Multi-VM Isolation and Threat Detection System in Virtualized Environments that provides secure isolation and continuous monitoring of multiple virtual machines running on a shared host. The system is designed to detect suspicious activities such as unauthorized access, abnormal network traffic.

Contribution: I developed the complete security system, including VM isolation and access-control mechanisms, hypervisor-level monitoring, threat detection using intrusion detection techniques, real-time security alerts, activity logging, and a monitoring dashboard for administrators.

Model Description: The system uses three main components — Virtual Machines, Security Monitoring Module, and Alert & Reporting Module. Each VM is isolated using access-control and virtualization security policies. Suspicious activities are identified by analyzing network traffic, system logs, and resource behavior, and an alert is generated when a predefined security threshold is exceeded.

Student Signature

Guide Signature